THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

031/2C PHYSICS 2C

(Actual Practical)

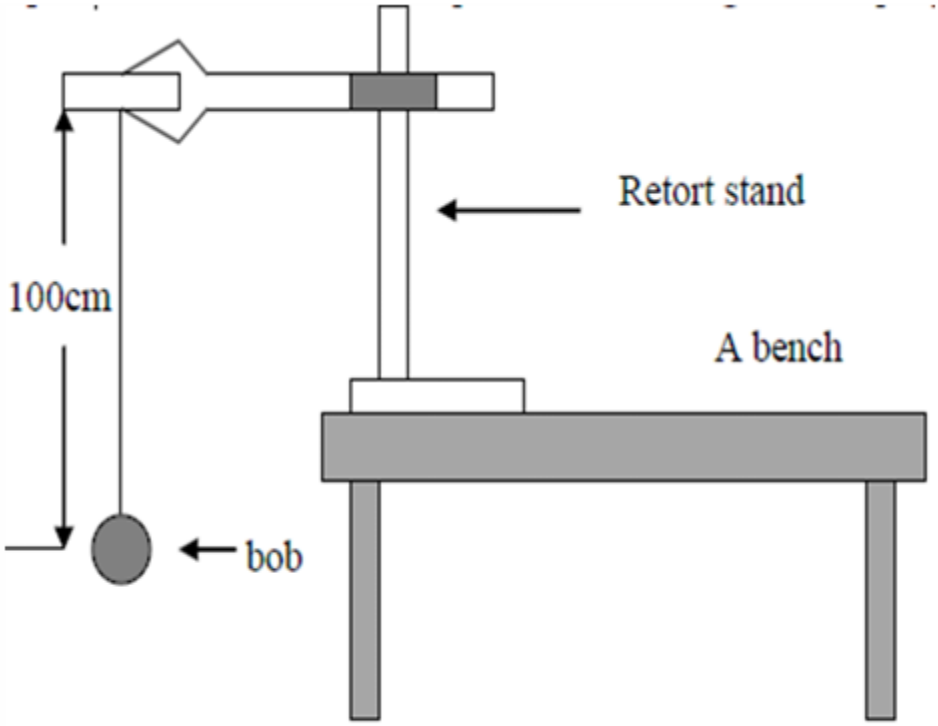
TIME: 2:30 HOURS YEAR: 2025

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.

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1. The aim of this experiment is to determine the acceleration due to gravity. You are provided the following apparatus; retort stand, stop watch, cotton thread, pendulum bob, cork pads, meter rule, arrange the apparatus as shown bellow



(a) Set the apparatus given as shown in figure 1 bellow

(b) Measure length L =100cm, pull the pendulum bob aside and release it so as to swing in small amplitude

(c) Find the time t, for 20 oscillations

(d) Find the periodic time T(s) and T^2 (S^2)

(e) Repeat the procedures for L=80cm, 60cm, 40cm, and 20cm.

(f) Tabulate your results of L, t, T and T^2

(g) Plot a graph of T^2 against L.

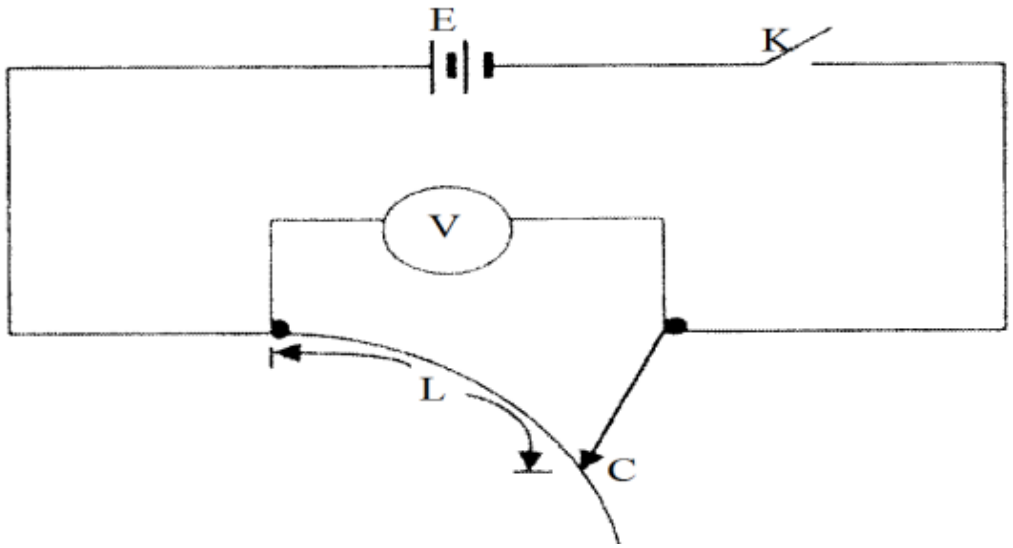
(h) Find the slope of the graph

(i) Using the relation T-$2\pi\sqrt{L/g}$ show how you can obtain acceleration due to gravity

(j) Calculate the value for the acceleration due to gravity, g

(k) Write two sources of errors

2. You are provided with the following apparatus; two dry cell, switch, a voltmeter, connecting wires, a wire and crocodile clip
Set up a circuit as shown in figure 2.



(a) Use micrometer screw gauge measure a dimeter of the wire Q and then find its area

(b) Measure the length of a wire L=100 CM and clip it by using crocodile clip, record the voltmeter reading V.

(c) Repeat the procedures in a above with crocodile clip at L=80cm, 60cm, 40cm and 20cm and record the corresponding voltmeter reading in each case

(d) Tabulate the values of L, V, 1/L and 1/V

(e) Plot a graph of $\frac{1}{\mathrm{V}}(V^{-1}$)against $\frac{1}{\mathrm{L}}(m^{-1})$

(f) Determine the slope of S

(g) Find the value of V for which $\frac{1}{\mathrm{L}}$ =0 and state its physical meaning

(h) Determine the value of resistivity of the wire given that

$$\frac{1}{\mathrm{V}} = \frac{\mathrm{A}}{\mathrm{PE}}\left(\frac{1}{\mathrm{L}}\right) + \frac{1}{\mathrm{E}}$$